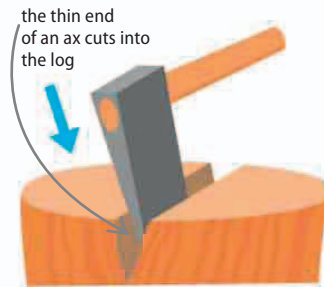


Machines

Physics allows us to build machines that harness forces and the energy they transfer to do useful work. A machine is a device that carries out a task by changing forces in some way. Machines need not be complex; in fact, a piece of high-tech machinery, such as a robot or an engine, is really a series of much simpler machines working together. Simple machines include levers, wheels, screws, ramps, and pulleys. Machines make work easier by converting small forces into big ones.

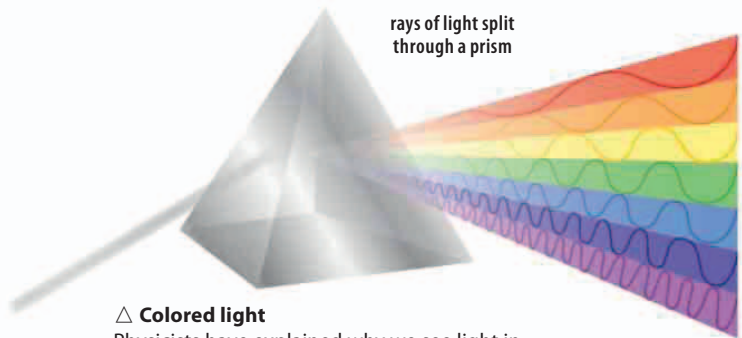


◁ Focusing force

Even the blade of an ax is a machine. The force pressing on the wide end of the wedge-shaped blade is focused into the sharp edge so it slices through solid objects.

Radiation

People are often confused by the term “radiation,” thinking it refers to the dangerous particles blasted out by nuclear reactions. However, in physics the word “radiation” normally refers to waves of light, heat, and other invisible rays that travel across the Universe. Together they form the electromagnetic spectrum, which is made up of mostly familiar types of radiation. As well as light, the spectrum includes radio waves, gamma rays, ultraviolet light, infrared (or heat), and X-rays. These are all examples of electromagnetic waves.

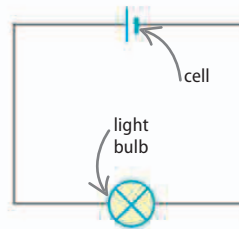


△ Colored light

Physicists have explained why we see light in different colors. Waves of red light are longer (and have less energy) than waves of violet light. All the other colors are in between.

Electricity

Thanks to physicists researching sparks and magnetic forces, most machines are powered by electric currents. This process began in ancient times, when early scientists examined magnetic, iron-rich stones that stuck to each other. Over the centuries, it was discovered that magnetism and electricity are linked—an area of physics called electromagnetism. This field also involves atomic structure and where radiation comes from.

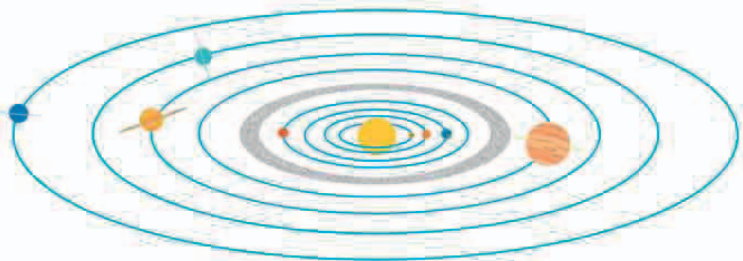


◁ Electric circuits

Electricity can be put to work using a circuit of different components. For example, a light bulb turns electric current into light when a cell is connected.

Astronomy

In many ways, astronomy was the first science of all, because ancient people saw patterns in the movement of the planets, Moon, and Sun. Modern astronomy still involves stargazing, but high-tech telescopes are used to gather light and other radiation from farther out in space than ever before. The laws of physics discovered on Earth work in just the same way on the other side of the Universe. Therefore, astronomers can use their knowledge to understand the many different objects they see out in space—and even figure out how the Universe came into existence.



our Solar System

△ Meet the neighbors

Observations of the eight planets in our Solar System have taught us much about our own world. Astronomers are now searching for Earth-like planets around more distant stars.